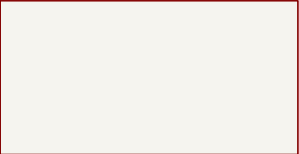


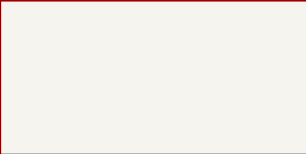
# Data Center RDIMM DDR5 Tester

HyperRAM



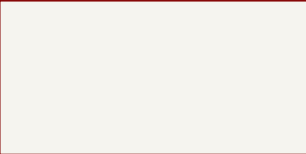
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DDR5



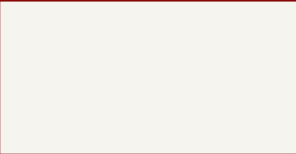
File: DDR5.kicad\_sch

Interfaces



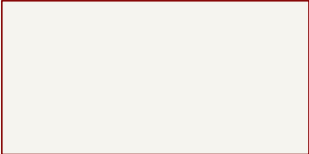
File: interfaces.kicad\_sch

Ethernet



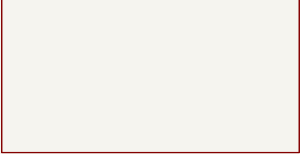
File: ethernet.kicad\_sch

Supply



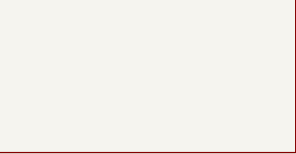
File: supply.kicad\_sch

Config SPI flash



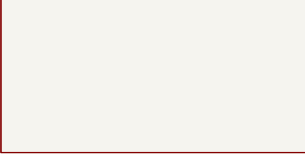
File: config-spi.kicad\_sch

FPGA power



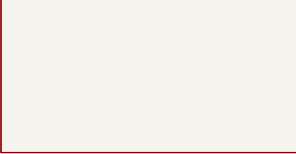
File: fpga-power.kicad\_sch

FPGA banks 12-15



File: fpga-banks-12-15.kicad\_sch

FPGA banks 16-34



File: fpga-banks-16-34.kicad\_sch



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Sheet: /  
File: data-center-rdim--ddr5-tester.kicad\_sch

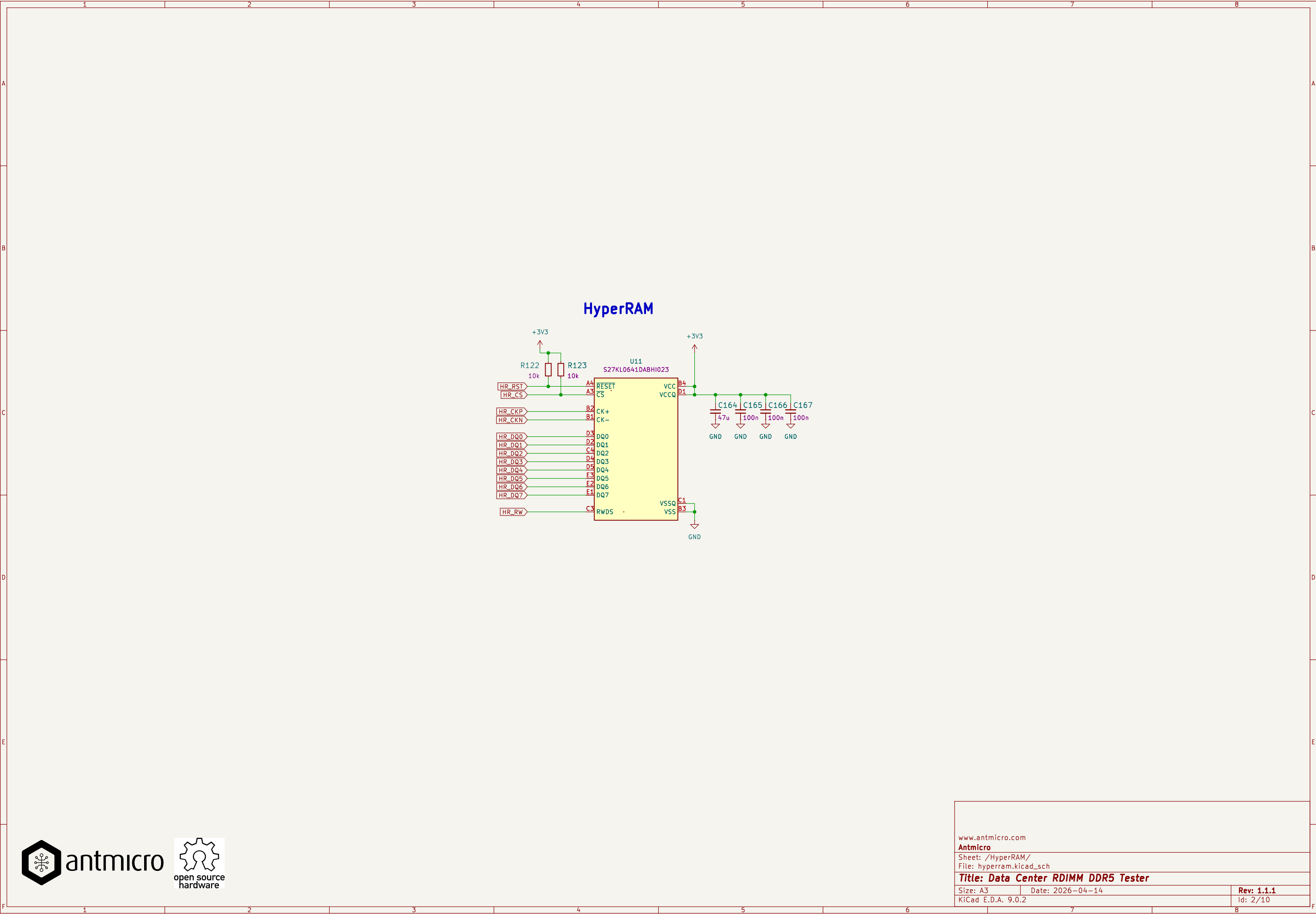
Title: Data Center RDIMM DDR5 Tester

Size: A3  
KiCad E.D.A. 9.0.2

Date: 2026-04-14

Rev: 1.1.1

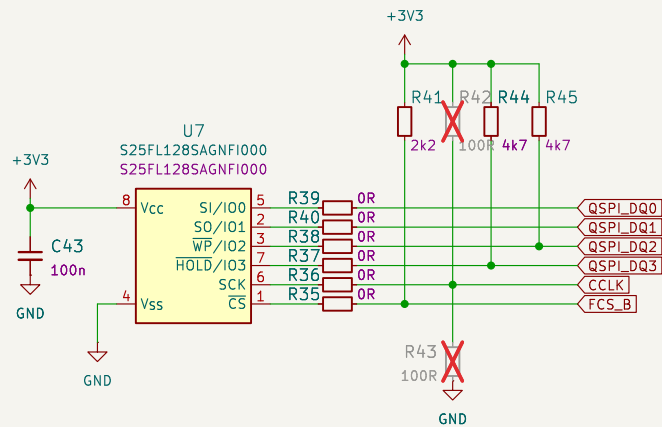
Id: 1/10



Master SPI Quad (x4) configuration scheme

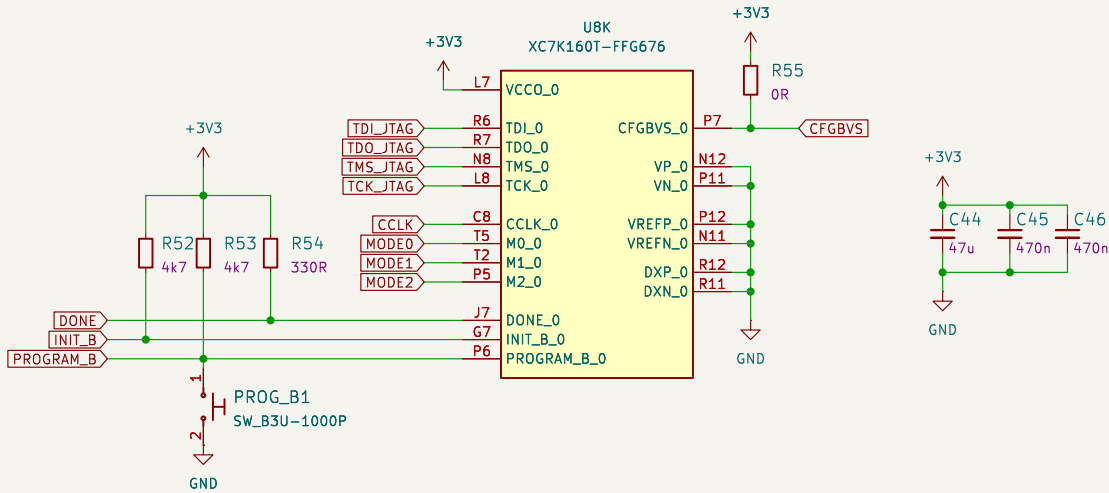
Follows Figure 2-14 7 Series FPGAs Configuration User Guide  
UG470 (v1.13.1)

(Q)SPI flash



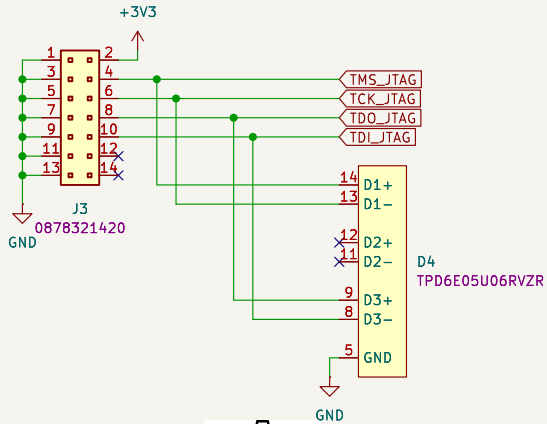
FPGA BANK 0

TOD0: verify after FPGA swap

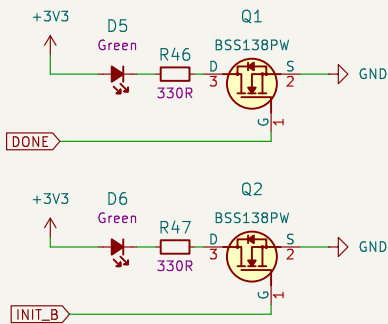


JTAG Connector

Compatible with Xilinx Platform Cable

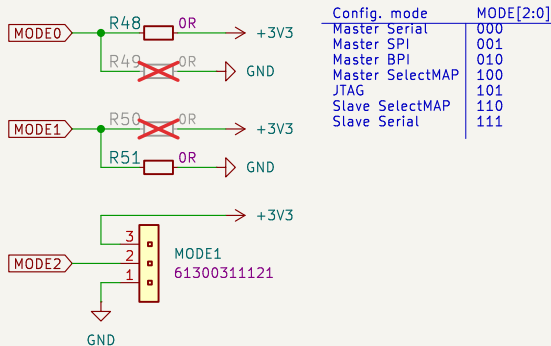


STATUS LEDs



Configuration Modes

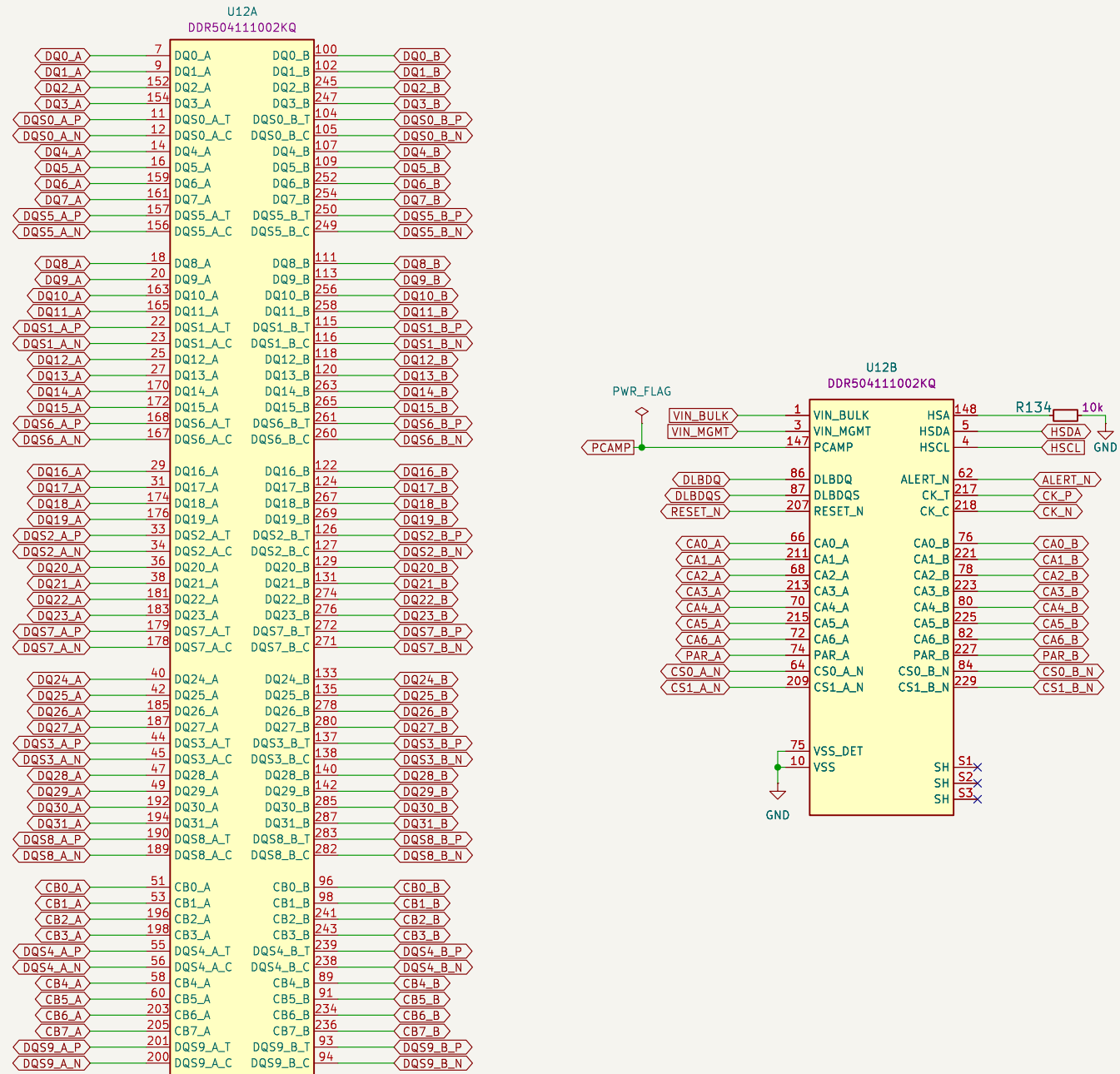
For details, see UG470 p. 21

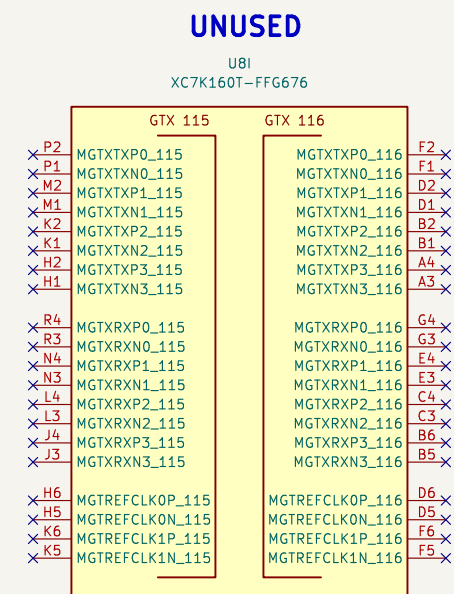
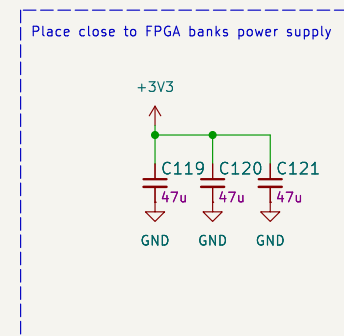
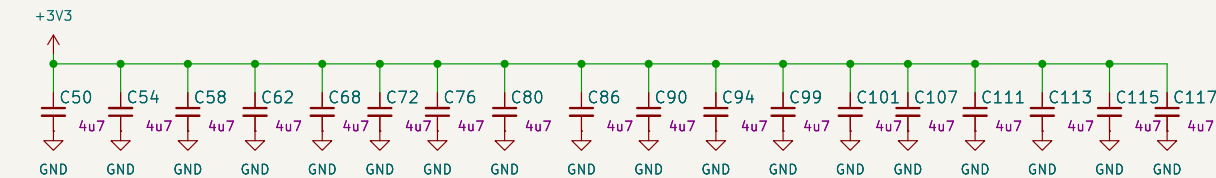


Probes

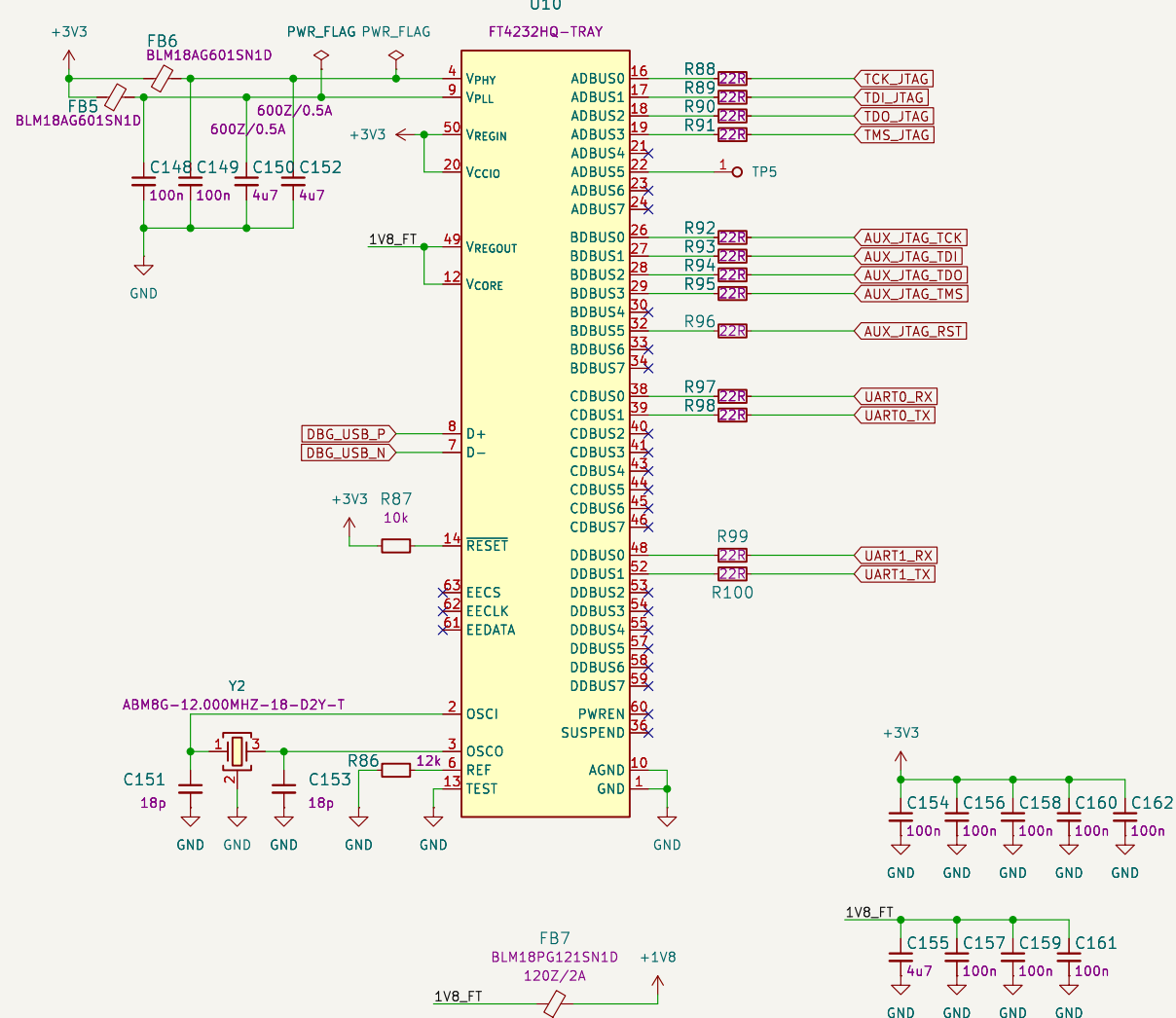


DDR5 RDIMM connector

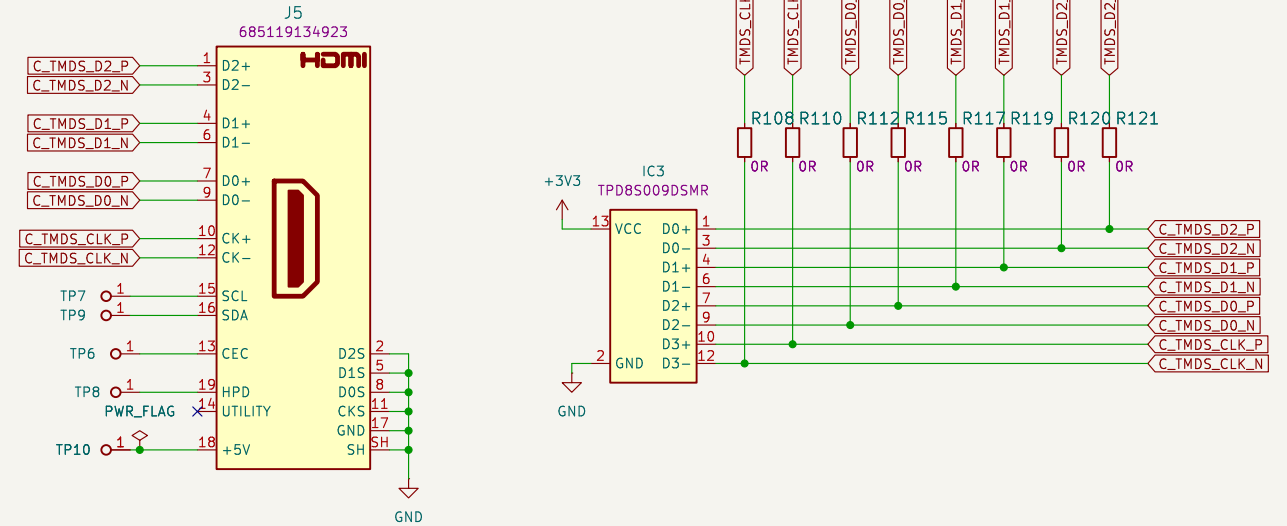




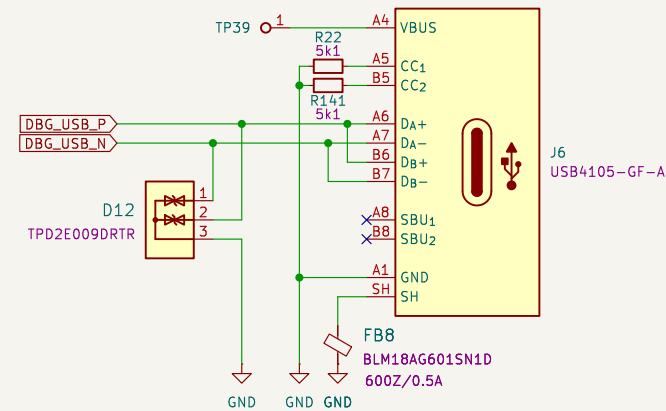
## Debug UART



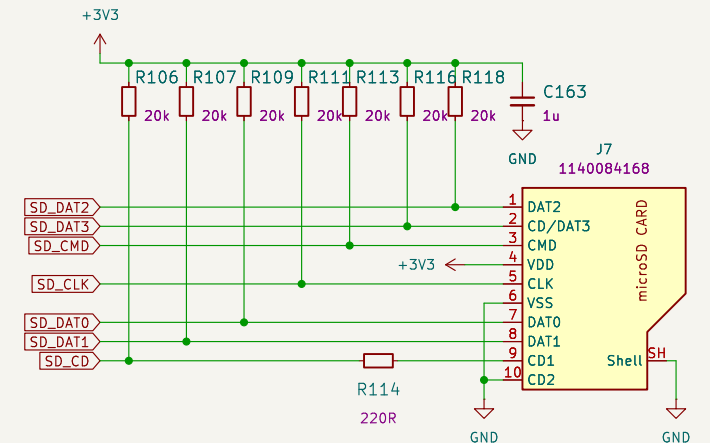
## HDMI



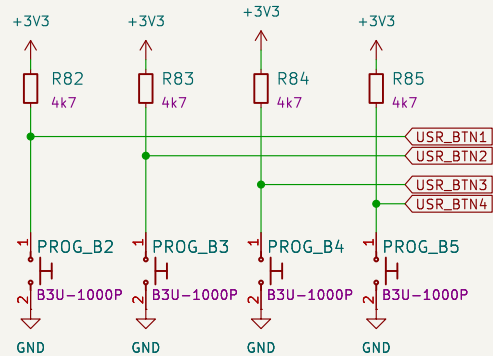
## Debug USB



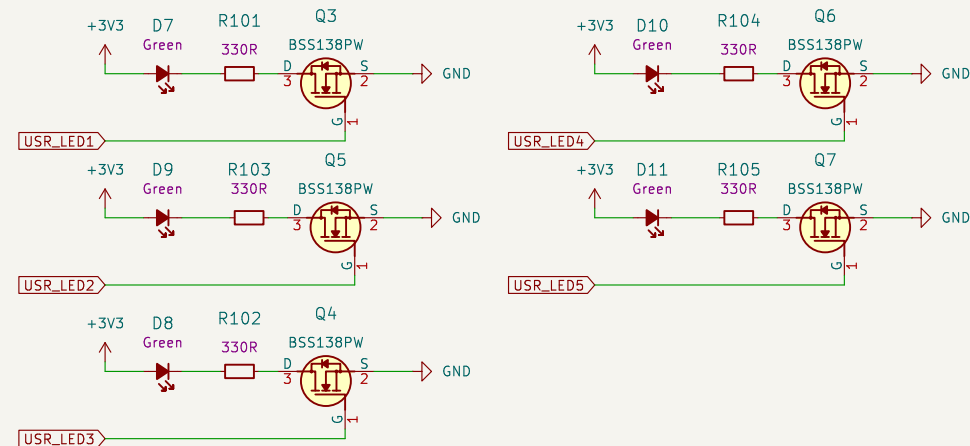
## SD card



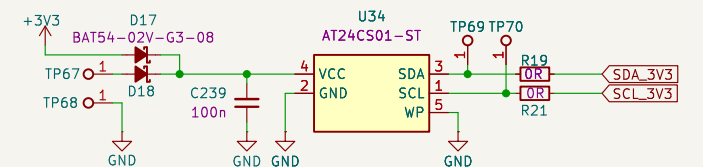
## User buttons



## User LEDs



## UID EEPROM



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Sheet: /Interfaces/

File: interfaces.kicad\_sch

Title: Data Center RDIMM DDR5 Tester

Size: A3

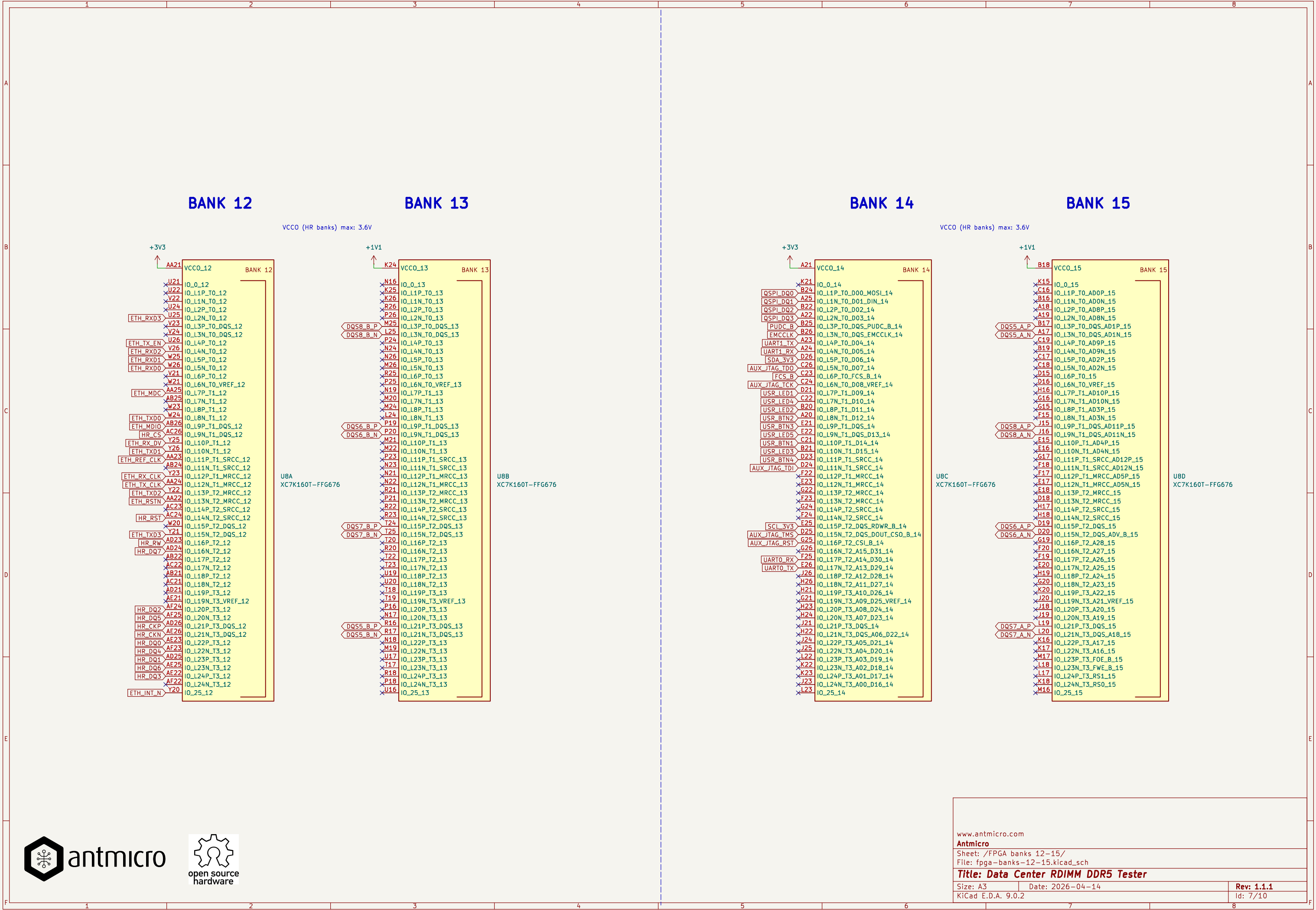
Date: 2026-04-14

Rev: 1.1.1

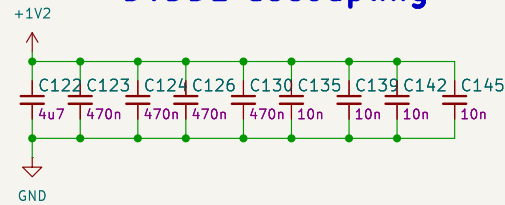
KiCad E.D.A. 9.0.2

Id: 6/10

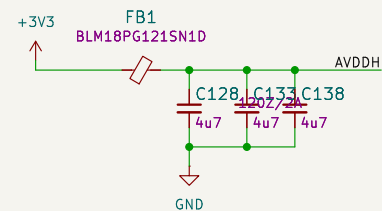




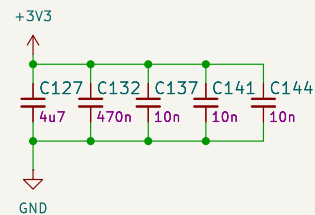
## DVDDL decoupling



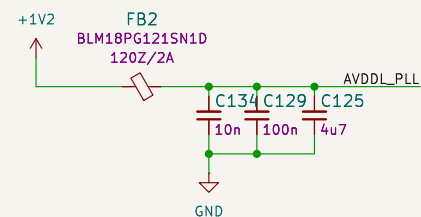
## AVDDH decoupling



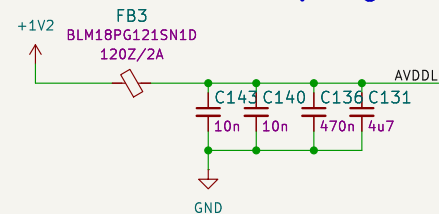
## DVDDH decoupling



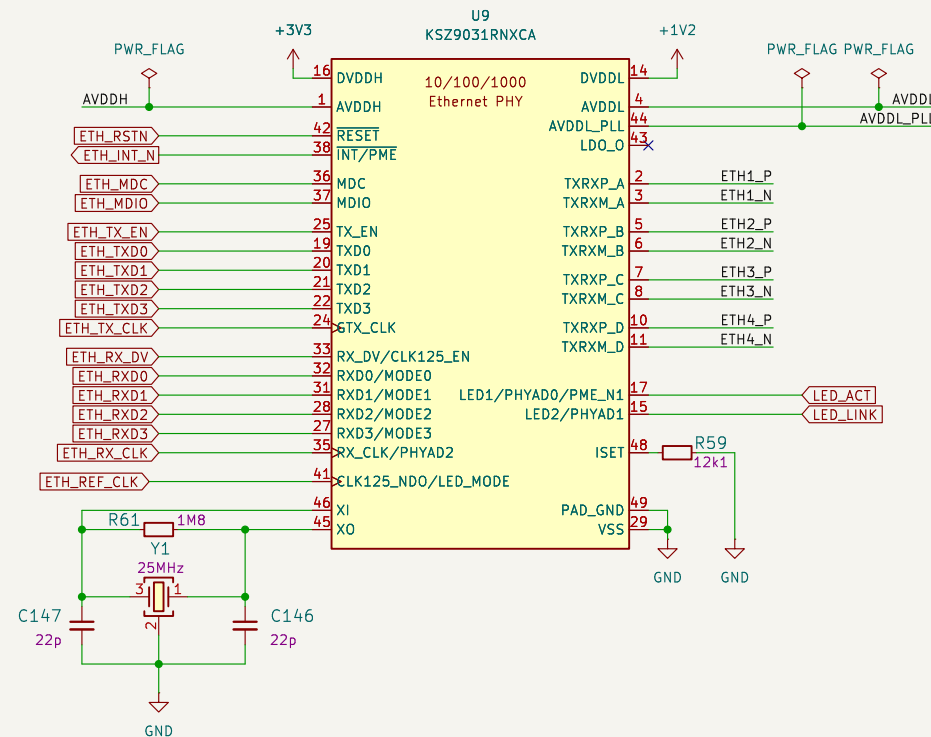
## AVDDL\_PLL decoupling



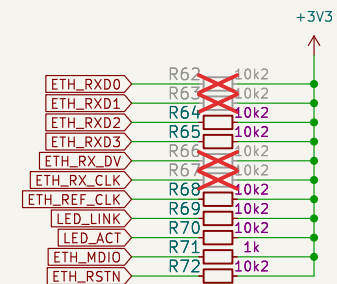
## AVDDL decoupling



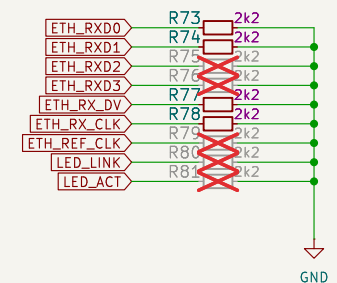
## PHY



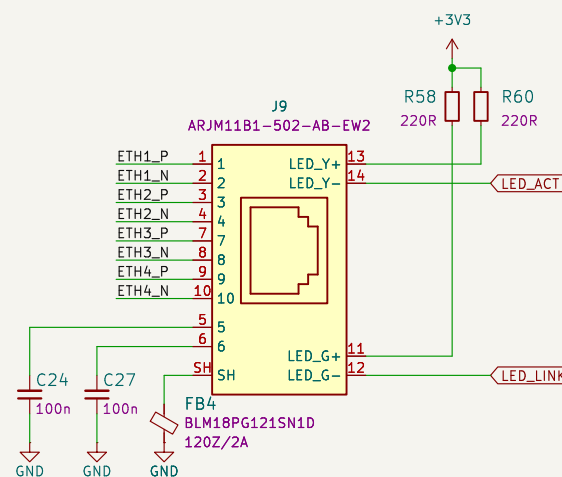
## Pull up resistors



## Pull down resistors



## RJ45 Connector



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Sheet: /Ethernet/

File: ethernet.kicad\_sch

**Title: Data Center RDIMM DDR5 Tester**

Size: A3

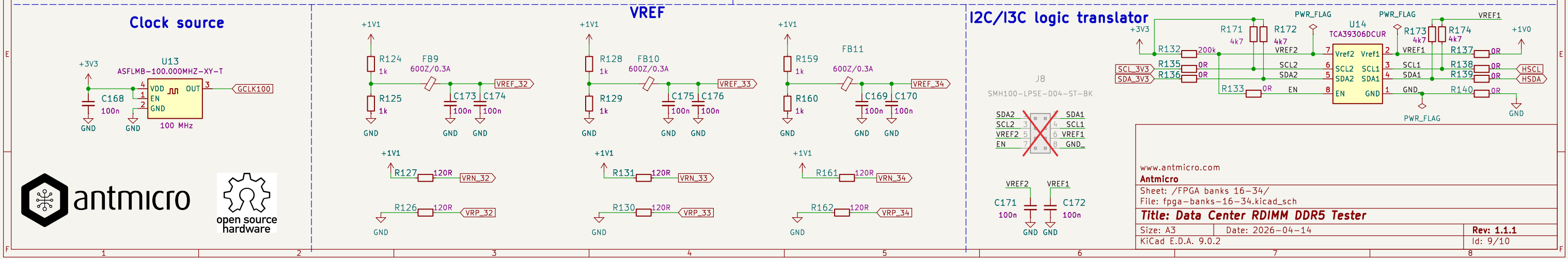
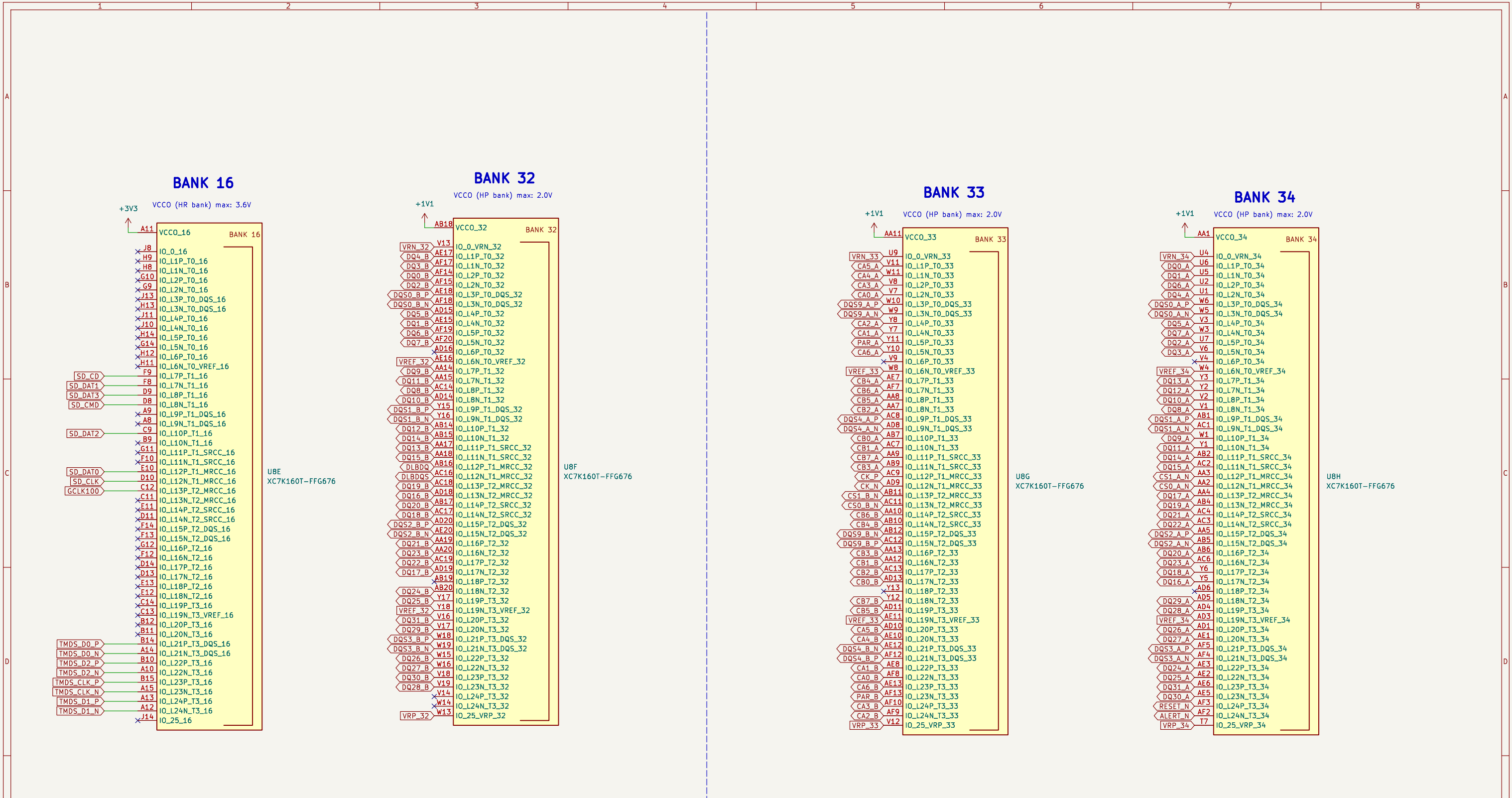
Date: 2026-04-14

Rev: 1.1.1

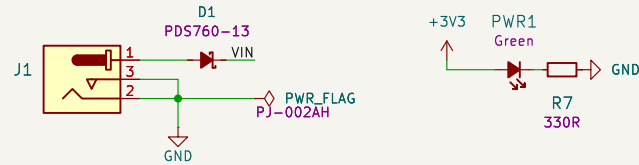
KiCad E.D.A. 9.0.2

Id: 8/10

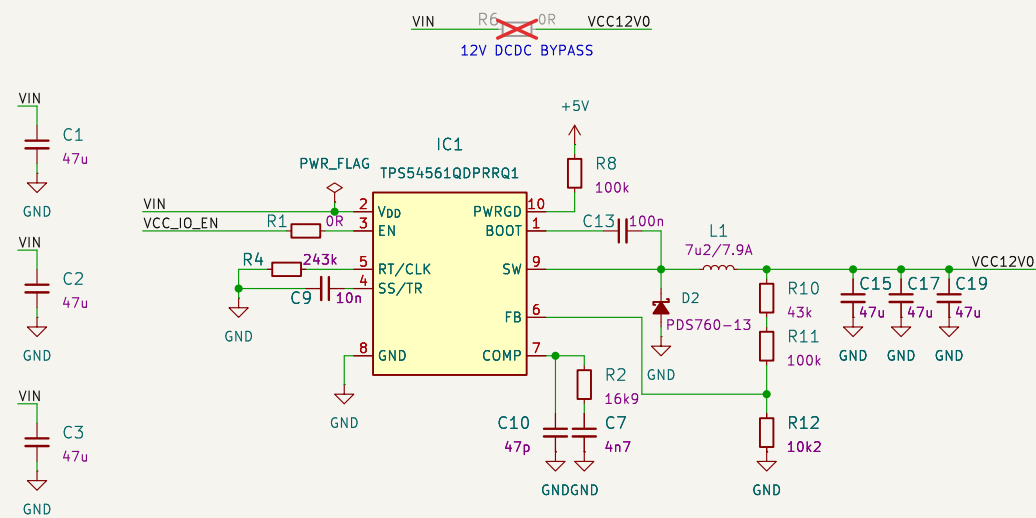




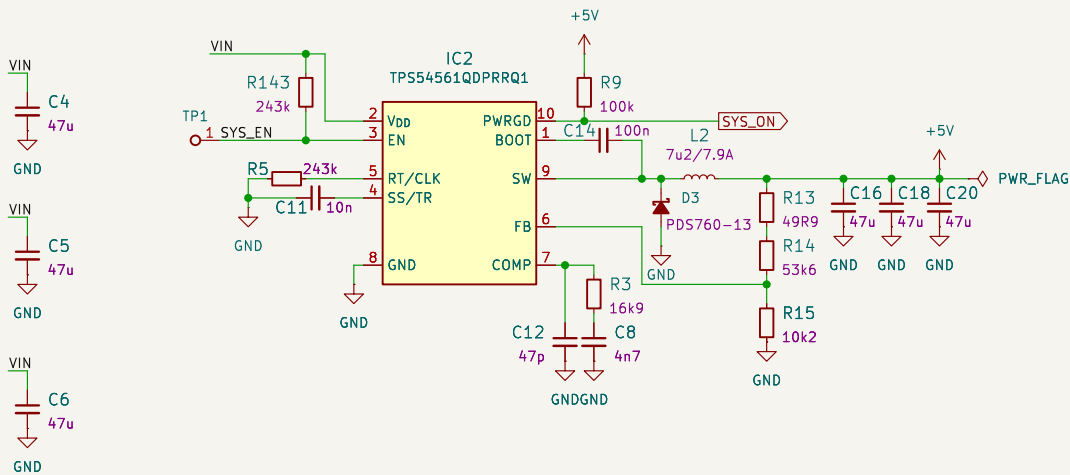
## Input power connector



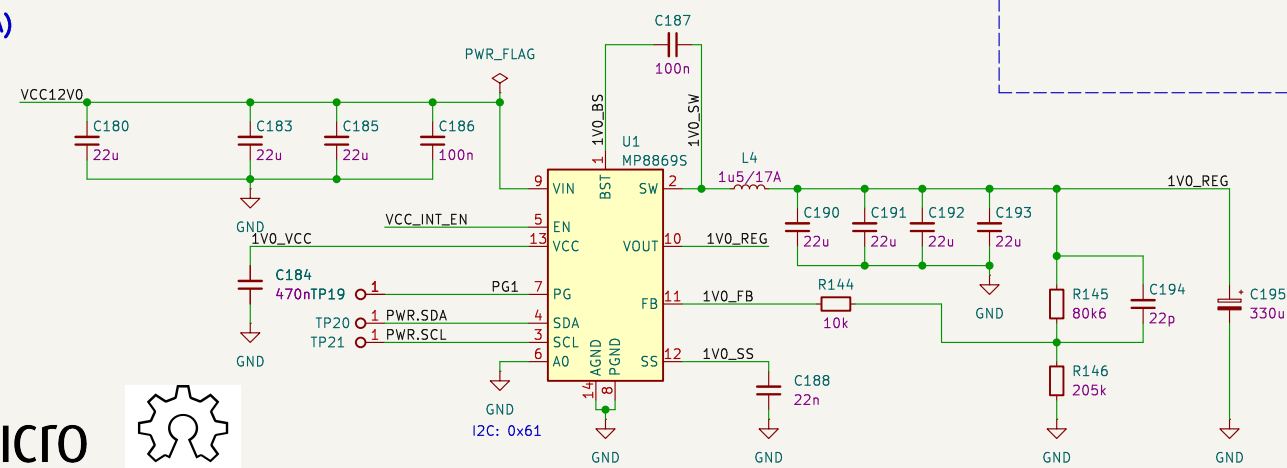
## Main supply (12V 5A)



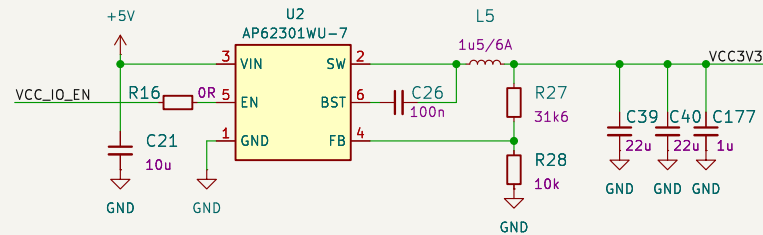
## 5V0 supply



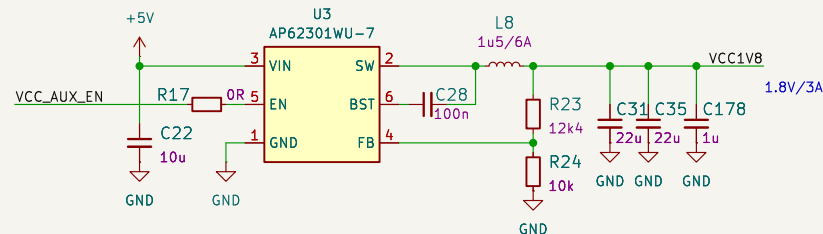
## VCCINT (1.0V 12A)



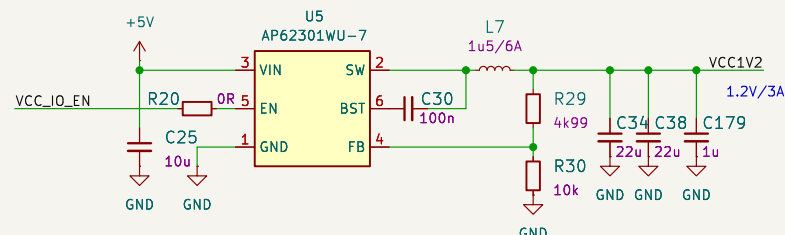
## 3V3 supply (3A)



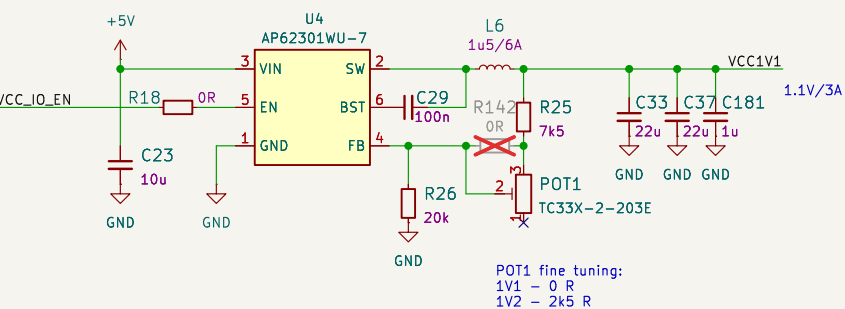
## 1V8 supply



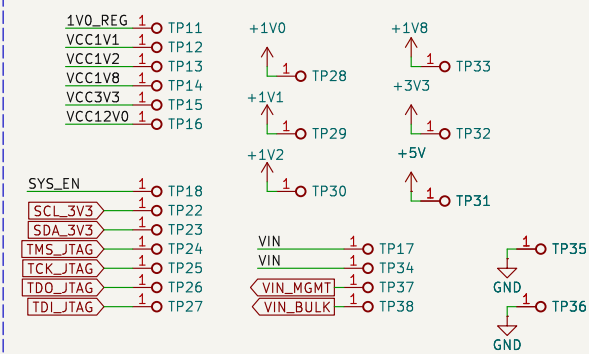
## 1V2 supply



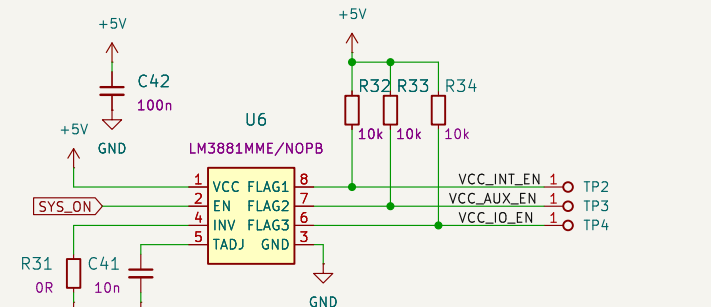
## 1V1 supply



## ICT Test Points

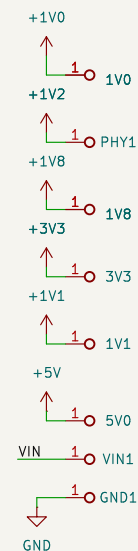


## Power sequencer

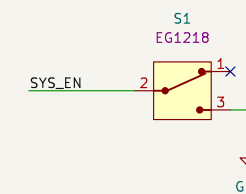


STEP1 - VCCINT (1.0V) for FPGA  
STEP2 - VCCAUX (1.8V, 2.5V, 1.2V) for FPGA and DDR  
STEP3 - VCCIO (3.3V, 1.2V, 0.6V) for FPGA, PHY and DDR

## Probes



## Power switch



## Optional FAN connector

